4B. Advanced Data Modelling and Form Design

Aim:

To extend the Gym Membership application using advanced fields, a trainer lookup, a family-member subform, media uploads, calculated values, an automatic membership identifier and a QR code.

Definition:

Advanced form design supports related records, repeating data, files, media, calculated values and system-generated identifiers. Lookup fields connect records between forms, while subforms collect multiple related rows within a parent record.

Advanced Field Design:

Field / Component	Type	Purpose
Assigned Trainer	Lookup	Reference a Trainers record; display its name and specialization.
Family Members Enrolled	Subform	Store up to five family-member rows with name, age and relationship.
Member Photo / Medical Certificate	Image / File Upload	Capture supporting images and documents.
Current BMI	Decimal	Store a supplied BMI value; no BMI calculation is defined.
Fitness Social Profile	URL	Store a web link.
Fitness Goal Video / Health Voice Note	Video / Audio	Capture supporting media.
Membership Fee Paid / Referral Discount	Currency / Percentage	Store fee and discount inputs.
Total Amount Payable	Formula	Calculate the discounted amount from fee and discount.
Gym Rules Display / Member Signature	Rich Text / Signature	Present rules and capture a signature.
Membership ID / Formatted ID	Auto Number / Formula	Generate a sequence and a formatted membership reference.
Member Access QR Code	QR Code	Encode the formatted membership identifier.
Enrollment Submitted On	Date-Time	Store a timestamp initialized through validation logic.

Data Relationships:

The Membership Enrollment form links to the Trainers form through Assigned_Trainer_Lookup, whose values reference Trainers.ID. One trainer may be selected by several membership records. Family_Members_Enrolled stores repeating family details within an enrollment, with a maximum of five rows.

Procedure:

1. Open the Gym Membership application in edit mode. Create or review the Trainers form with Trainer Name and Specialization.

2. Configure Assigned Trainer Lookup in Membership Enrollment to reference Trainers records. Display the trainer name and specialization.

3. Add the Family Members Enrolled subform with Family Member Name, Family Member Age and Relationship; set a maximum of five rows.

4. Add image, file-upload, decimal, URL, video and audio fields in suitable sections. Configure the medical-document field to allow PDF, JPG and PNG files.

5. Add currency and percentage fields for the fee and referral discount. Configure the Total Amount Payable formula.

6. Add rich text for gym rules, a mandatory signature field and a date-time field for enrollment submission.

7. Create the automatic Membership ID, the formatted identifier formula and a QR Code field that encodes the formatted identifier.

8. Save the form, enter demonstration records and inspect the report for calculated amounts, formatted identifiers and QR codes.

9. Check trainer selection, subform rows, uploaded files, signature capture and formula outputs using suitable test inputs.

Calculation and Identifier Logic:

Total Amount Payable = Fee Paid x (1 - Referral Discount Percent / 100). Null fee and discount values are treated as zero in the defined formula.

The formatted identifier combines GYM-MEM-, the year taken from the enrollment timestamp (or current time when absent), and the automatic number padded to six digits. The QR code encodes this identifier; it does not by itself implement entry authorization.

Output - Trainer Assignment and Media Fields:

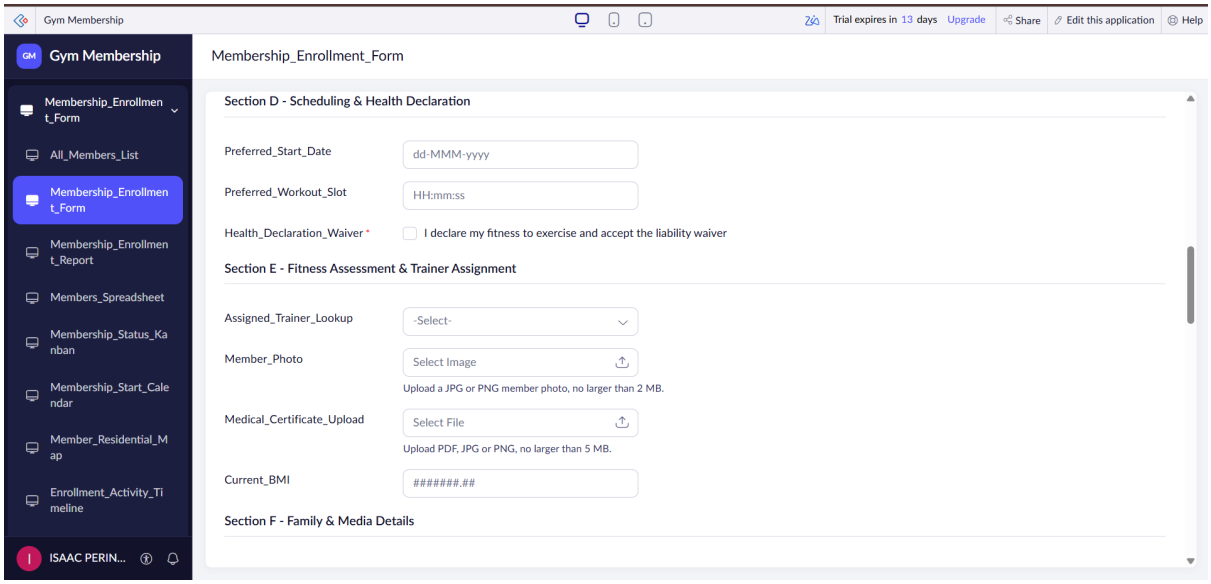


Figure 1. Trainer lookup, member photo, medical-certificate upload and decimal BMI field.

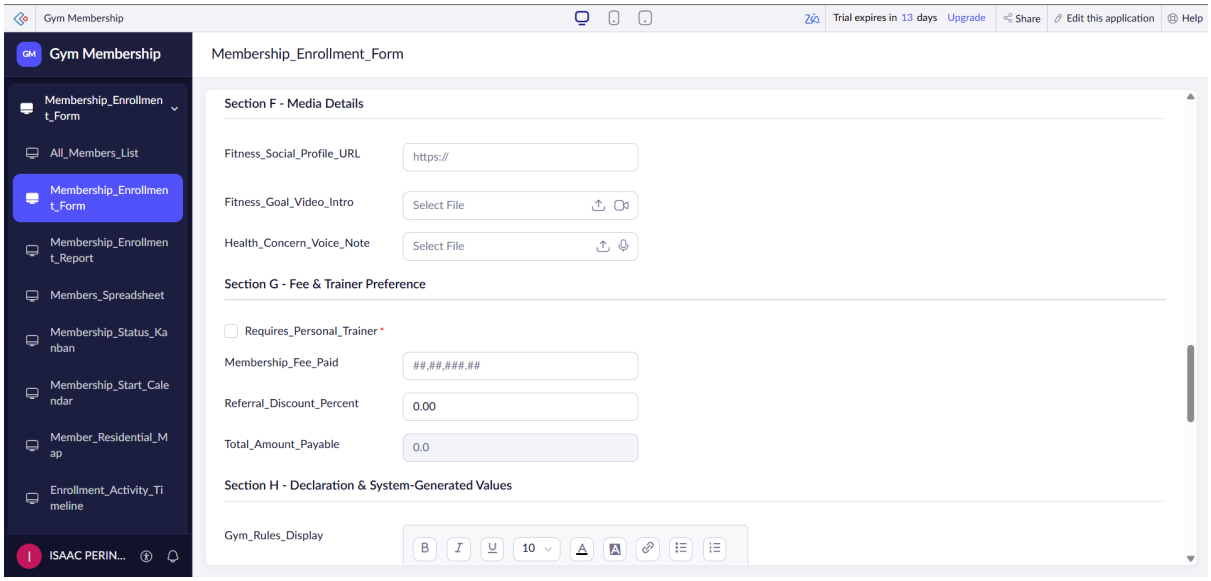


Figure 2. URL, video, audio, fee, referral discount and calculated amount controls.

Observation:

The form groups advanced inputs into fitness assessment, media and fee sections. Upload controls and a calculated amount field are visible. Upload-size messages shown beside the fields are guidance; the export alone does not confirm that those size limits are enforced.

Output - Signature, Identifiers and QR Codes:

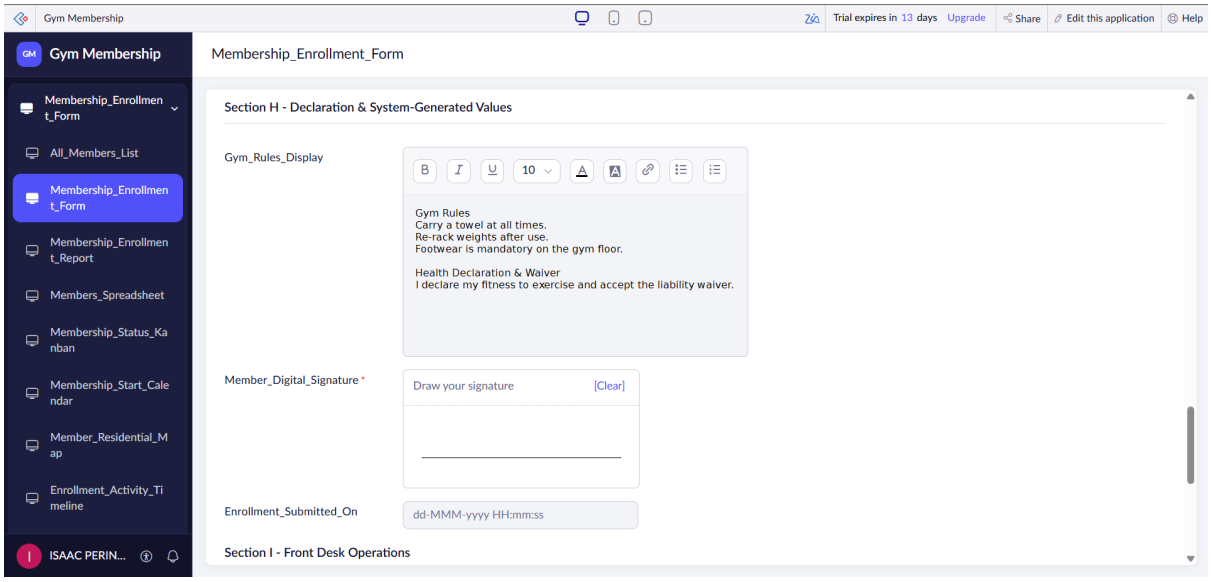


Figure 3. Gym rules, digital-signature input and enrollment date-time field.

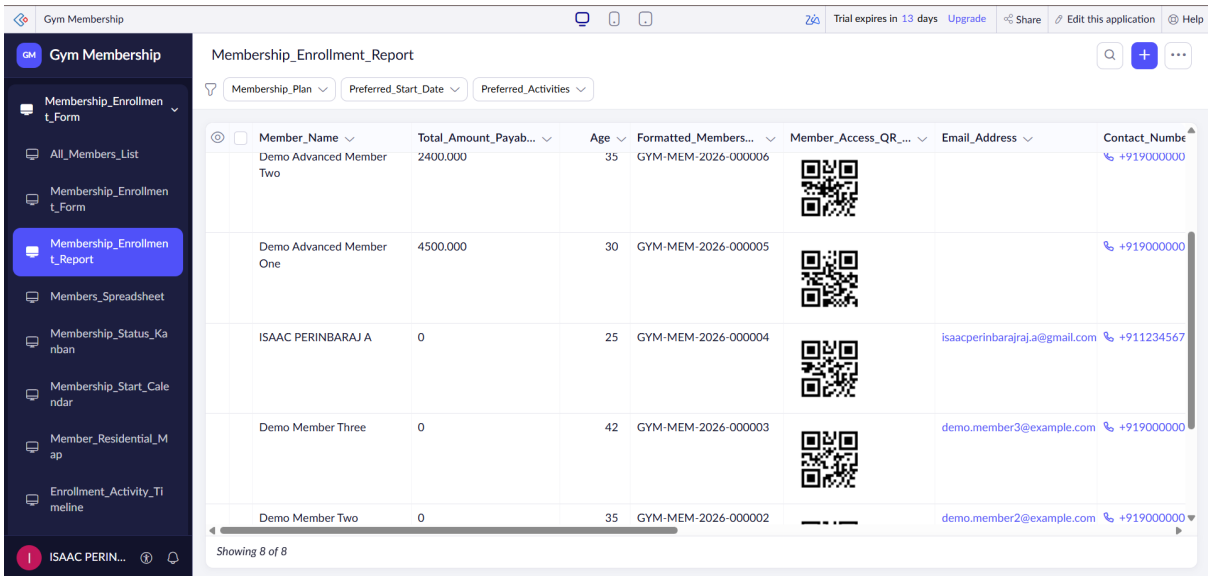


Figure 4. Saved report records showing calculated amounts, formatted membership IDs and QR codes.

Result:

Advanced form design was implemented in the Gym Membership application through lookup and subform definitions, media fields, calculated amounts and system-generated identifiers. The supplied form screens show advanced input controls, while the report demonstrates saved formatted IDs and QR codes. Trainer lookup selection, subform entry and file-upload behavior require their own test evidence to confirm successful operation.